

LAYAN KAUSHIK

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EDUCATION

Duke University, Durham, NC, USA

Aug 2022 – May 2024

M.S. Economics and Computation

GPA: 3.91/4.0

Courses: Theory and Algorithm of Machine Learning, Probabilistic Machine Learning, Elements of Deep Learning, Design and Analysis of Algorithms, Advanced Econometrics, Numerical Analysis, Graph Network Analysis, Algorithmic Game Theory.

Indian Institute of Technology, Roorkee, India

Jul 2016 – May 2020

B.Tech. Civil Engineering, First Division with Distinction

GPA: 8.726/10

Ranked 12 in a class of 121 students.

Courses: Computer Programming and Numerical Analysis, Engineering Mathematics, Probability and Statistics, Data Mining, Linear Optimization, Graduate Econometrics.

HONORS AND AWARDS

Dean's Research Award for Master's Students, Duke University

2023

U.S. Army Corps of Engineers Institute for Water Resources Research Grant, Washington DC

2023

Duke Graduate Working Group on Global Issues Award, Duke University

2023

Duke Economics Master's Scholar Award, Duke University

2022-24

K. C. Mahindra Scholarship for Post Graduate Studies Abroad, K. C. Mahindra

2022

Annual Excellence Award, IIT Roorkee Heritage Foundation

2018 & 2019

Chief Minister's Scholarship, Government of Assam, Assam, India

2018

IASc-INSANA Summer Research Fellowship, Indian Academy of Sciences

2018

IIT Roorkee ENCORE Award, IIT Roorkee

2017

NEC Merit Scholarship, Directorate of Technical Education, Government of Assam

2016

Top 3 in State Mathematics Olympiad, Homi Bhabha Centre for Science Education, India

2014

WORK EXPERIENCE

Economist Apprentice, Amazon, Seattle, US

Jun 2024 - Present

- Working alongside **Professor Alexandre Mas** and **Principal Economist Dr. Mingyu Chen** at the intersection of economics and statistics—specializing in survey methodology, measurement science, and causal inference frameworks.

Product Data Scientist, Mastercard, India

Aug 2020 - Jun 2022

- Worked as Product Data Scientist and Data Lead on Mastercard's Geo Insights platform, applying statistical and spatial analysis to generate reliable economic and demographic insights and guide market opportunity identification.

Field Engineer Trainee, Schlumberger (Rebranded as SLB), India

May 2019 - Jul 2019

- Worked as a Field Engineer Trainee in the upstream Oil & Gas division.

PUBLICATIONS

“Refining FEMA’s NFIP V2 Open Flood Insurance Claims Dataset: Comprehensive Enhancements through Data Augmentation, Feature Engineering, Anomaly Detection and Multiple Imputation”

American Geophysical Union (AGU) Conference 2023, ([Publication Link](#))

Enhanced flood insurance claims dataset through advanced data augmentation and feature engineering using Python geospatial libraries. Developed outlier detection models to identify water-depth anomalies and addressed critical dataset inconsistencies including missing data and measurement discrepancies, improving reliability for flood risk assessment and insurance policy analysis.

WORKING PAPERS

“SurveyBalancer: Statistical Survey Reweighting Package”

Amazon, January 2025 - Present

Developing an internal Python package for correcting non-response bias in survey data through machine learning-based

propensity score estimation and advanced statistical reweighting methods. Implementing leave-features-out cross-validation with distribution matching techniques and conducting extensive validation through statistical tests. Package being adopted by Amazon's benchmarking and Preview teams to raise the technical bar for survey research across Alexa, supporting product launch decisions for millions of users.

"A Large-Scale Test of Difference-in-Differences and Common Practices with Experimental Data"

Amazon, June 2025 - Present

Conducting systematic validation of Difference-in-Differences causal inference methods using approximately 500 large-scale experiments from Alexa, comparing DiD estimates against experimental ground truth across varying assumption violations and treatment effect heterogeneity patterns. Developing frameworks to assess DiD estimator performance under realistic violations of parallel trends, examining robustness across different pre-treatment period lengths, treatment timing patterns, and covariate adjustment strategies.

"Strategy-Proof Mechanism Design for Drop-Down Tournaments: Theory, Algorithms, and Implementation"

Duke University, June 2024 - Present

Investigating incentive compatibility in multi-stage tournaments where players may strategically underperform to secure easier future matchups. Proposing a strategy-proof reward system using a scaled bracket-based incentive structure and dynamic Elo-style rating updates; proving incentive compatibility mathematically and validating the model through Monte Carlo simulations across varied player strategies. Applying theoretical insights to real-world tournament formats including Pickleball and UEFA Champions League, revealing design flaws and suggesting incentive-aligned alternatives.

RESEARCH ASSISTANT EXPERIENCE

Applying NLP to Uncover Insights in Influencer Marketing, Guided by Prof. Tong Guo, The Fuqua School of Business, Duke University

Jan 2024 - May 2024

- Analyzed 1.29 million ADHD-related TikTok videos using NLP and unsupervised ML, classifying each as Ad or non-Ad.
- Applied topic modeling to identify behavioral patterns in user engagement, focusing on uncovering prevalent themes and trends within ADHD-related discussions on TikTok.
- Skills: Python, Machine Learning, Natural Language Processing, Web Scrapping, Topic Modeling

Treatment Effect Estimation With Measurement Error in Covariates and Repeat Measurements, Guided by Prof. Michael Pollmann, Duke University

May 2023 - Jan 2024

- Derived and implemented an unbiased objective method for higher-order polynomials under homoskedastic measurement error and multiple mismeasured variables from "Identification and Estimation of Polynomial Errors-in-Variables Models" by Hausman et al.
- Collaboratively developed an R package, optimizing existing codes for time and space efficiency, and executed extensive large dataset simulations on Duke's Computer Cluster.
- Implemented Kenichi Nagasawa's "Treatment Effect Estimation with Noisy Conditioning Variables" in R, demonstrating the model's versatility across various datasets.
- Crafted and executed cross-validation code in R for CoCoLasso regularization in linear regression and linear inverse probability weighted estimation, addressing datasets with measurement errors.
- Skills: R, C++, Causal Inference, Machine Learning

Addressing User Activity Bias in Bipartite Graph Ranking, Guided by Prof. Xiaobai Sun, Duke University

Sep 2023 - Dec 2023

- Enhanced the BiRank algorithm for bipartite graph rankings in recommendation systems, incorporating user activity bias mitigation through a novel regularization technique.
- Developed a unique fairness metric, combining activity rank balance and Gini coefficient, to evaluate the improved algorithm's effectiveness in diverse graph scenarios.
- Executed extensive experimental analysis on synthetic datasets, demonstrating significant fairness improvements in rankings, particularly for power-law distributed graphs.
- Skills: Advanced algorithm design, Experimental Design

Ride Sharing and Tipping Behavior in Chicago from 2018 to 2023, Guided by Prof. Jian Pei, Duke University *Jan 2023 - May 2023*

- Analyzed 300M+ Chicago ride-sharing records (2018-2023) using regression, time-series analysis, and hypothesis tests; identified statistically significant patterns, including a 33% ride drop post-pandemic and 1.5% tipping increment, reflecting socio-economic trends.
- Utilized Python for large dataset management; implemented chunk-based data processing, batch processing, and dynamic sampling to optimize data tasks.
- Skills: Time-Series Analysis, Regression, Hypothesis Testing

Spatial Distribution of Moisture Content in Cement Mortar Structures using Radial Basis Functions, Guided by Prof. Bishwajit Bhattacharjee, IIT Delhi *May 2018 - Aug 2018*

- Conducted extensive research on structural engineering, sustainable environment, basic electrical science, and interpolation methods to measure moisture content in cement mortar required to obtain durability and service life of the structure.
- Incorporated an efficient, accurate, and reliable method to estimate the moisture content in laboratory-synthesized cement concrete blocks using data obtained from experiments conducted during the research.
- Utilized interpolation techniques like the Gaussian and inverse multiquadric radial basis function method in the estimation function to determine moisture content for which data could not be generated and verified with the theoretical calculations.
- Skills: Python (Libraries: NumPy, Matplotlib, SciPy)

TEACHING EXPERIENCE

Graduate TA, CS230: Discrete Math for CS, Duke University *Jun 2023 - Aug 2023*
Responsibilities include conducting office hours, developing recitation slides, and delivering lectures.

Graduate TA, CS216: Everything Data, Duke University *Jan 2023 - May 2023*
Responsibilities include holding office hours, managing undergraduate TAs, creating assignments, and automating grading.

Undergraduate TA, Computer Programming in C++, IIT Roorkee *Jan 2018- May 2018*
Held office hours and lab sessions with over 40 students.

Undergraduate TA, Solid Mechanics, IIT Roorkee *Jul 2017 - Nov 2017*
Held office hours and tutorial sessions with over 40 students.

SKILLS

Programming Languages: Python (scikit-learn, PyTorch, pandas, statsmodels, econml, geopandas), SQL, R, C++

ML Frameworks: PyTorch, JAX

Software Packages: Stata, Tableau, MS Excel, LaTeX

Spoken Languages: English (fluent), Assamese (native speaker), Hindi (fluent)

VOLUNTEER WORK AND OTHER ACTIVITIES

Assistant Badminton Coach, Duke University *Aug 2022 - May 2023*
Provide regular training and coaching advice to players affiliated with Duke Badminton Club.

Student Mentor, Student Mentorship Program, IIT Roorkee *Jul 2018 - May 2019*
Mentored a batch of 7 freshmen to help with their transition from high school to university.

Professional Player Badminton, India *Jan 2010 - May 2020*
Represented the Indian state of Assam in various Sub-Junior and Junior National Championships in India. Was awarded Best Badminton Player of IIT Roorkee for three consecutive years 2017, 2018, and 2019.